

Question number	Answer	Mark
2(a)	The only correct answer is A. <i>B is incorrect because this is not a method of counting bacteria</i> <i>C is incorrect because colonies are counted not weighed</i> <i>D is incorrect because colonies are counted on agar. Turbidity would not work on agar.</i>	(1)

Question number	Answer	Additional guidance	Mark
2(b)	An explanation that includes the following points: - attach lid to dish to ensure {lid does not fall off / bacteria do not enter / bacteria do not leave / no contamination} (1) - not completely sealed {so that conditions remain aerobic / prevent the growth of anaerobic bacteria} (1)	ACCEPT a description for both marking points e.g. sellotape used in places / sellotape not used all the way round ACCEPT a description of how to attach e.g. tape up the petri dish, use sellotape to seal petri dish ACCEPT leave gaps so there is oxygen NB leave air holes in the sealing = this mark	(2)

Question number	Answer	Mark
2(c)(i)	The only correct answer is C. <i>A is incorrect because lag comes before exponential and death comes at the end</i> <i>B is incorrect because exponential comes between lag and death</i> <i>D is incorrect because lag comes before exponential and stationary comes after</i>	(1)

Question number	Answer	Mark
2(c)(ii)	The only correct answer is D. <i>A is incorrect because all four are correct</i> <i>B is incorrect because all four are correct</i> <i>C is incorrect because all four are correct</i>	(1)

Question number	Answer	Additional guidance	Mark
2(c)(iii)	$\log_{10} N_t$ and $\log_{10} N_0$ values read from graph and subtracted (1) 2.49 (1)	6.5 – 2 / 4.5 IGNORE 'log' before the 6.5 and 2 if they clearly haven't used the log value ecf if log of logs have been taken	(2)